



HK IMO Training 2023-2024 Problem Set 20 Suggested Solutions



Problem 77 (Iran MO 2016 3rd Round N1). Let S be a set of positive integers with at least two elements. Let $P(x)$ be a polynomial with integer coefficients. Suppose that for any $a, b \in S$, we have $a + b \in S$ and $\gcd(P(a), P(b)) = 1$. Prove that $P(x)$ is a constant polynomial.

Solution. Let a and b be two distinct elements in S , and let $d = (a, b)$. It is well-known that there exists a positive integer N such that for any $n > N$, the integer dn can be represented in the form $ka + \ell b$ for some positive integers k and ℓ . As

$$ka + \ell b = ((((((a + b) + b) + \cdots) + b) + a) + \cdots) + a,$$

this implies all such dn belong to S .

If $P(x)$ is not a constant polynomial, consider any integer $n > N$ such that $P(dn) \neq \pm 1$. Let p be a prime divisor of $P(dn)$. Then $P(d(n + p)) \equiv P(dn) \equiv 0 \pmod{p}$. This implies $p \mid (P(d(n + p)), P(dn))$, but $d(n + p), dn \in S$. This is a contradiction. Therefore, $P(x)$ must be a constant polynomial. $\square$

Problem 78 (Iran MO 2020 3rd Round A2). Let $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n, c_1, c_2, \dots, c_n$ be real numbers. Prove that

$$\sum_{\text{cyc}} \sqrt{\sum_{i=1}^n (3a_i - b_i - c_i)^2} \geq \sum_{\text{cyc}} \sqrt{\sum_{i=1}^n a_i^2}.$$

Solution. Let $\mathbf{a} = (a_1, a_2, \dots, a_n)$, $\mathbf{b} = (b_1, b_2, \dots, b_n)$ and $\mathbf{c} = (c_1, c_2, \dots, c_n)$. The desired inequality means

$$|3\mathbf{a} - \mathbf{b} - \mathbf{c}| + |-\mathbf{a} + 3\mathbf{b} - \mathbf{c}| + |-\mathbf{a} - \mathbf{b} + 3\mathbf{c}| \geq |\mathbf{a}| + |\mathbf{b}| + |\mathbf{c}|.$$

Indeed, by the triangle inequality, we have

$$2|3\mathbf{a} - \mathbf{b} - \mathbf{c}| + |-\mathbf{a} + 3\mathbf{b} - \mathbf{c}| + |-\mathbf{a} - \mathbf{b} + 3\mathbf{c}| \geq |2(3\mathbf{a} - \mathbf{b} - \mathbf{c}) + (-\mathbf{a} + 3\mathbf{b} - \mathbf{c}) + (-\mathbf{a} - \mathbf{b} + 3\mathbf{c})| = 4|\mathbf{a}|,$$

$$|3\mathbf{a} - \mathbf{b} - \mathbf{c}| + 2|-\mathbf{a} + 3\mathbf{b} - \mathbf{c}| + |-\mathbf{a} - \mathbf{b} + 3\mathbf{c}| \geq 4|\mathbf{b}|,$$

$$|3\mathbf{a} - \mathbf{b} - \mathbf{c}| + |-\mathbf{a} + 3\mathbf{b} - \mathbf{c}| + 2|-\mathbf{a} - \mathbf{b} + 3\mathbf{c}| \geq 4|\mathbf{c}|.$$

Adding these and dividing by 4, we obtain the desired inequality. $\square$

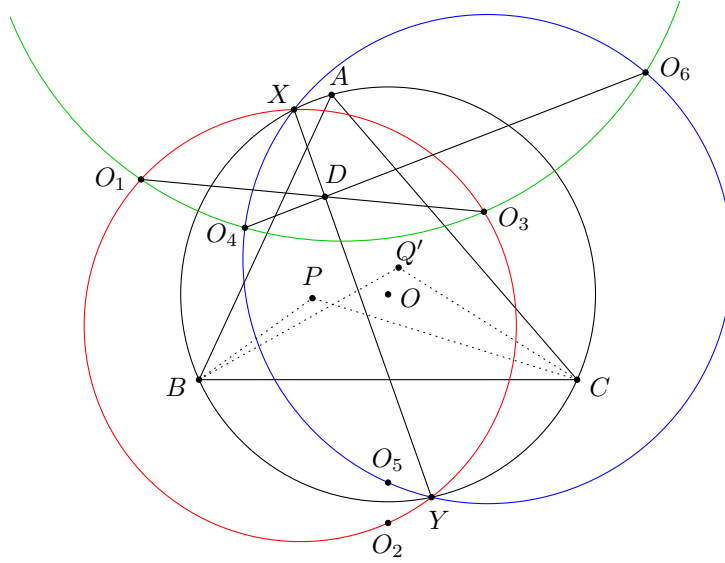
Problem 79 (RMM 2020 Problem 3 simplified). Let $n \geq 3$. We colour all edges of a complete graph K_n using n colours such that all colours are used and the edges of the same colour form a single odd cycle. Prove that there exists an odd cycle such that all edges in this cycle have different colours.

Solution. Let $v_1, v_2, \dots, v_m$ be an odd cycle of colour C_1 .

- If $m = 3$, there must be another vertex w different from the v_i 's as there are at least $m = 3$ colours. If there exist v_i, v_j such that $v_i w$ and $v_j w$ are of different colours (which must be different from C_1 by the given condition), then v_i, v_j, w is a cycle of length 3 such that all edges have different colours. Thus, the edges $v_1 w, v_2 w, v_3 w$ are of the same colour C_2 . However, this is impossible as the edges of colour C_2 cannot form a cycle (since there cannot be a vertex in a cycle incident with 3 edges).
- If $m = 5$, we may assume all edges $v_i v_{i+2}$ are of colour C_2 (where all indices are taken modulo 5), since otherwise we can always pick two edges $v_i v_{i+2}$ and $v_{i+2} v_{i+4}$ of different colours, and the cycle v_i, v_{i+2}, v_{i+4} satisfies the condition. Now, there must be another vertex w different from the v_i 's as there are at least $m = 5$ colours. As above, we may assume $v_1 w, v_2 w, v_3 w, v_4 w, v_5 w$ are of the same colour, which again yields a contradiction.
- If $m \geq 7$, we may assume $v_1 v_5, v_2 v_5, v_3 v_5$ are of the same colour (otherwise, either v_1, v_2, v_5 or v_2, v_3, v_5 satisfies the condition). But this is again a contradiction as above.

$\square$

Problem 80 (RMM Shortlist 2023 G3). Let P be a point inside $\triangle ABC$. Let O_1, O_2, O_3 be the circumcentres of $\triangle APB, \triangle BPC, \triangle CPA$ respectively. The circumcircles of $\triangle ABC$ and $\triangle O_1O_2O_3$ intersect at two points X and Y . Let Q be the reflection of P in the line XY . Prove that $\angle BAP = \angle CAQ$.



Solution. Let Q' be the isogonal conjugate of P with respect to $\triangle ABC$. It suffices to prove $Q' = Q$. Let O_4, O_5, O_6 be the circumcentres of $\triangle AQ'B, \triangle BQ'C, \triangle CQ'A$ respectively. Also, let O be the circumcentre of $\triangle ABC$.

Note that O, O_2, O_5 are collinear as they lie on the perpendicular bisector of BC . As

$$\begin{aligned}\angle OBO_2 &= \angle OBC + \angle CBO_2 = \left(\frac{\pi}{2} - \angle CAB\right) + \left(\frac{\pi}{2} - \angle BPC\right) = \angle BAC - \angle BPC \\ &= \angle ABP + \angle PCA = \angle Q'BC + \angle BCQ' = \angle BQ'C = \angle BO_5O,\end{aligned}$$

we have $OO_5 \times OO_2 = OB^2 = R^2$, where R is the circumradius of $\triangle ABC$. Similarly, we have $OO_4 \times OO_1 = OO_6 \times OO_3 = R^2$. Therefore, O_1, O_4, O_3, O_6 are concyclic.

Consider the inversion about (ABC) . From above, the points in each of the pairs (O_1, O_4) , (O_2, O_5) and (O_3, O_6) are mapped to each other. Thus, $(O_1O_2O_3)$ is mapped to $(O_4O_5O_6)$. As X and Y lie on (ABC) , they are invariants. Thus, X and Y also lie on $(O_4O_5O_6)$. This implies O_1O_3, O_4O_6, XY are concurrent at the radical centre D of $(O_1O_2O_3)$, $(O_4O_5O_6)$ and $(O_1O_4O_3O_6)$. Note that O_1O_3 is the perpendicular bisector of AP , while O_4O_6 is the perpendicular bisector of AQ' . Therefore, D is the circumcentre of $\triangle APQ'$. Analogously, the circumcentre E of $\triangle BPQ'$ also lies on XY . As $D \neq E$, we see that the line DE , or the line XY , is the perpendicular bisector of PQ' . This proves $Q' = Q$. $\square$